FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

BIOLOGY

Past Paper 2025 • Paper Code: 1HA-I 25006 — 30061 (Version B) / 32061 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

BIOLOGY HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 30061 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. When energy flows from one trophic level to the next, the amount of energy lost is about:

(A) 10%
(B) 50%
(C) 90%
(D) 100%

Q2. The part involved in detoxification of drugs is: — [A diagram of a cell organelle with labelled parts P, Q, S, R is printed next to the question in the source scan]

(A) P
(B) Q
(C) R
(D) S

Q3. The main function of progesterone in human female reproductive system is:

(A) Ovulation
(B) Follicle development
(C) Endometrium development
(D) Implantation of embryo

Q4. When RNA polymerase binds to promoter for transcription, the unit which detaches form it is:

(A) α (Alpha)
(B) β (Beta)
(C) γ (Gamma)
(D) θ (Theta)

Q5. Which one of the following sequence of reactions is correct regarding Calvin cycle?

(A) 3PGA→1-3BPG→ G3P
(B) 3PGA→G3P→RuP
(C) G3P→RuBP→3PGA
(D) G3P→RuP→3PGA

Q6. The stem cells which can differentiate into few cell types are:

(A) Totipotent
(B) Unipotent
(C) Pluripotent
(D) Oligopotent

Q7. The protein present in tendons and ligaments is:

(A) Casein
(B) Keratin
(C) Elastin
(D) Fibrinogen

Q8. The compound malonate is competitive inhibitor of:

(A) Fumarate
(B) Succinate
(C) Succinic dehydrogenase
(D) Histidine decarboxylase

Q9. The condition in which centromere of a chromosome is located slightly away from the center is termed as:

(A) Acrocentric
(B) Sub metacentric
(C) Telocentric
(D) Metacentric

Q10. Which fungus is poisonous?

(A) Agaricus
(B) Morchella
(C) Aspergillus
(D) Amanita

Q11. Hepatitis–D is caused by:

(A) Bacteria
(B) Virus
(C) Prions
(D) Viroids

Q12. Bark is produced by:

(A) Cork cambium
(B) Lateral meristem
(C) Apical meristem
(D) Intercalary meristem

Q13. The sex linked traits which are more common in males than female are:

(A) Autosomal recessive
(B) Y– linked dominant
(C) X– linked dominant
(D) X– linked recessive

Q14. The organism having Chlorophyll – a, Phycocyanin and Heterocyst may be a/an:

(A) Bacterium
(B) Cyanobacterium
(C) Fungus
(D) Alga

Q15. Which of the following structure is common to all chordate embryo?

(A) Placenta
(B) Notochord
(C) Vertebral column
(D) Lungs

Q16. Identify the trait which is NOT controlled by polygenes?

(A) Intelligence
(B) Height
(C) Skin color
(D) Tongue rolling

Q17. Which of the following is an example of homologous organs?

(A) Wing of an insect & a bird
(B) Forelimb of human & wing of a bat
(C) Wing of an insect & a bat
(D) Gills of fish & lungs of mammals

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 25006 (B)

Q2. (Marks (14 x 3 = 42)) Q.2 Answers the following parts briefly.

(i) [03] How does Golgi complex play role in Cell secretion and Cell division? — OR (03) — Compare the characteristics of Cofactors of enzymes with one example of each.

(ii) [03] Write any three main features of Bryophytes. — OR (03) — Enlist any three physical methods used to control bacteria with their mechanisms.

(iii) [03] Justify the statement "Water is a universal solvent". — OR (03) — Endospore formation is a mechanism in bacteria to survive under unfavorable conditions. Justify.

(iv) [03] List the Accessory glands associated with male reproductive system with their respective locations. — OR (1+2) — What is incomplete dominance? Explain with a suitable example.

(v) [1+2] Why and how cyclic Photophosphorylation occurs in light reactions? — OR (03) — Write steps of lytic cycle in bacteriophage.

(vi) [1+2] What is mycorrhiza? Differentiate between its types. — OR (1+2) — What is point mutation? How it causes sickle cell anaemia?

(vii) [03] Compare apoplast and symplast pathways involved in uptake of water by roots in plants. — OR (03) — Molecular biology provides evidence of evolution. Justify the statement with reference to any three examples.

(viii) [1.5+1.5] Write about de-nitrification and assimilation processes in nitrogen cycle. — OR (2+1) — Write about the structure and functions of ovary and oviduct in female reproductive system.

(ix) [1x3] What is Y– linked inheritance? How these traits are transmitted? Also give two examples. — OR (03) — Name and briefly explain Growth Phases of bacteria. (Any three)

(x) [03] Complete the table with reference to enzymes: [a: Enzyme — , Mechanism of action: Break large molecules with water, Example — ; b: Enzyme: Lyases, Mechanism — , Example — ; c: Enzyme — , Mechanism — , Example: Cytochrome oxidase] — OR (2+1) — Coat color in Labrador retriever shows epistasis. If a black(BbEe) and a chocolate (bbEe) Labradors retrievers are crossed: a. Show genotypes and phenotypes of offspring(s). b. Calculate the probability of black and yellow coat color.

(xi) [03] Illustrate the conversion of pyruvic acid to acetyl CoA as link reaction between glycolysis and Krebs cycle. — OR (03) — Interpret the results of Meselson and Stahl experiment when they grew bacteria in N_{15} and then in N_{14} medium for tube 2, 3 and 4. — [A diagram of four centrifuge tubes (Control, 0 Min, 20 Min, 40 Min) is printed next to the question in the source scan]

(xii) [03] Write difference in flower, stem and seed of dicots and monocots. — OR (03) — Name any three enzymes with their functions involved in DNA replication.

(xiii) [03] Elaborate the endosymbiont theory regarding evolution of chloroplast and mitochondria. — OR (1x3) — How is maltose different from sucrose in glycosidic linkage, monosaccharide units and specific role?

(xiv) [03] What are the limitations of Lamarckism? — OR (03) — Write characteristics of angiosperms with reference to vascular tissues and alternation of generation.

> Note: Attempt the following questions.

Q3. (Marks 5+2) Q.3 Explain the steps of life cycle of Human Immunodeficiency Virus (HIV). Also draw its life cycle. — OR (07) — Explain different stages of xerarch succession on bare rock starting form lichens to flowering plants.

Q4. (Marks 2+2+1+2) Q.4 Explain translation process starting from initiation complex till termination. Also draw diagram of initiation complex and elongation steps. — OR (5+2) — How electrons move through electron transport chain till they reach final acceptor? Explain. Draw diagram of electron transport chain.

Q5. (Marks 4+2) Q.5 Describe the pathways involved in moving signals from outside to inside the cell with their diagrams. — OR (3+3) — Elaborate the structures and roles of steroids and prostaglandins as important group of lipids.

Q6. (Marks 06) Q.6 Explain the mechanism of photoperiodism. — OR (2x3) — Write three distinguishing characteristics and one example of each of the phylum Cnidaria Arthropoda and Mollusca.

VERSION D

BIOLOGY HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 32061 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. In tRNA molecule, the main function of middle loop is to:

(A) Recognize enzyme activity
(B) Bind codon on mRNA
(C) Recognize binding site on ribosome
(D) Attach amino acid to CCA

Q2. Prostaglandins are derivatives of:

(A) Amino acids
(B) Steroids
(C) Archidonic acid
(D) Nucleic acids

Q3. What is correct for protein signaling? [I: Water soluble, II: Binds receptor on plasma membrane, III: Lipophilic, IV: Do not require cAMP]

(A) I & II
(B) II & III
(C) III & IV
(D) I & IV

Q4. Which labelled part in nucleus is responsible for ribosome synthesis? — [A diagram of a nucleus with labelled parts P, S, Q, R is printed next to the question in the source scan]

(A) P
(B) Q
(C) R
(D) S

Q5. Which enzymes break specific covalent bonds without hydrolysis?

(A) Oxidoreductases
(B) Hydrolases
(C) Transferases
(D) Lyases

Q6. Viroids cause:

(A) Hepatitis A
(B) Hepatitis B
(C) Hepatitis C
(D) Hepatitis D

Q7. Type of flagella arrangement shown in bacteria is: — [A diagram of a bacterium with flagella is printed next to the question in the source scan]

(A) Lophotrichous
(B) Peritrichous
(C) Amphitrichous
(D) Monotrichous

Q8. Genetic material moves from donor to recipient bacterium in:

(A) Binary fission
(B) Conjugation
(C) Transduction
(D) Transformation

Q9. Mutualistic fungi which grows between root cells of plants and does not penetrate cells is:

(A) Lichens
(B) Endomycorrhiza
(C) Ectomycorrhiza
(D) Epiphytes

Q10. Growth movements shown by climbing plants with a supporting structure is:

(A) Phototropism
(B) Geotropism
(C) Thigmotropism
(D) Chemotropism

Q11. During unwinding of DNA double helix, the hydrogen bonds between base pairs are broken by enzyme:

(A) Helicase
(B) Primase
(C) Polymerase
(D) Gyrase

Q12. Radial symmetry and water vascular system are found in phylum:

(A) Hemichordata
(B) Echinodermata
(C) Chordata
(D) Vertebrata

Q13. Uterine lining in female starts breaking due to less concentration of:

(A) Progesterone
(B) Estrogen
(C) Luteinizing hormone
(D) FSH

Q14. A forest was completely destroyed by fire. After about hundred year, a new forest was evolved. The type of succession shown is:

(A) Primary and xerarch
(B) Secondary and xerarch
(C) Primary and hydrarch
(D) Secondary and hydrarch

Q15. Holandric traits show all of the following characteristics EXCEPT:

(A) Sons of affected father are affected
(B) Show Y-linked inheritance
(C) Porcupine man is an example
(D) Daughters of affected father are carriers

Q16. In TS of a dicot stem, which substances are expected to pass through labelled parts P and Q respectively? — [A diagram of TS of a stem with labelled parts P and Q is printed next to the question in the source scan]

(A) Water and sucrose
(B) Sucrose and amino acids
(C) Sucrose and water
(D) Water and amino acids

Q17. The structures which help bacteria to survive under unfavorable conditions include:

(A) Flagella and pili
(B) Pili and akinetes
(C) Endospores and plasmid
(D) Flagella and heterocyst

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 25006 (D)

Q2. (Marks (14 x 3 = 42)) Q.2 Answers the following parts briefly.

(i) [1x3] The diagram shows the section through testis: a) Identify the parts A & B. b) Write functions of C & D. c) What is the function of inhibin hormone in males? — [A diagram of testis section with labels A, B, C, D is printed next to the question in the source scan] — OR (03) — Write the major points of theory of natural selection. (Any three)

(ii) [03] Compare any three adaptations for osmotic adjustments in hydrophytes and mesophytes. — OR (1+2) — Why post–transcriptional modifications occur in eukaryotic mRNA? Summarize the changes which occur during mRNA processing.

(iii) [2+1] Write about the structure of prions. Enlist two diseases caused by them. — OR (03) — Write any three distinguishing characteristics of vascular plants.

(iv) [1x3] What are the homologous organs? Write an example. Which type of evolution is caused by them? — OR (1x3) — Write any three chemical methods with their mechanisms used to control bacteria.

(v) [03] Complete the table: [a) Group: Bryophytes, Character — , Example — ; b) Group — , Character: Seedless plants, Example — ; c) Group — , Character — , Example: Rose] — OR (1+2) — Name the type of pattern of sex determination in humans. Also write mechanism of sex determination in man.

(vi) [03] Write the difference between leaf, seed and root of monocots and dicots. — OR (03) — Write down any three roles of smooth endoplasmic reticulum in a cell.

(vii) [03] Peroxisomes are more abundant in liver cells. Justify. — OR (03) — Outline the process of non-cyclic photophosphorylation with the help of a labelled flow sheet.

(viii) [2+1] The diagram shows molecule of Lecithin: a) Identify parts A, B, C & D. b) Mention the polarity of E & F. — [A diagram of a Lecithin molecule with labels A–F is printed next to the question in the source scan] — OR (03) — Following is the diagram of HIV: a) Identify the parts A, B & C. b) Write the functions of protein gp40, gp120 and enzyme integrase in HIV. — [A diagram of HIV with labels A, B, C is printed next to the question in the source scan]

(ix) [03] Differentiate between absorption spectrum of Chlorophyll– a, b and Carotenoids. — OR (03) — How can a virus survive under unfavorable conditions outside the host?

(x) [03] Justify that O –ve persons are universal donors and AB +ve are universal recipients of blood. — OR (03) — How is photosynthesis in Cyanobacteria different from Bacteria? (any three differences)

(xi) [03] How are Starch and Glycogen different from each other although both are polysaccharides? — OR (03) — Why Lysosomes are also called suicidal bags?

(xii) [03] Enlist any three fungal diseases in humans with one symptom each. — OR (03) — Name the types of chromosomes on the basis of position of centromere mentioning its location in each type. (Any three)

(xiii) [03] Write about the structure and function of each of the three layers of uterus. — OR (03) — How evolution of Giraffe neck can be explained on the basis of Lamarckism?

(xiv) [03] Write down the sources of mutagens with their examples. (Any three) — OR (03) — Name any three main greenhouse gases and their sources of emission.

> Note: Attempt the following questions.

Q3. (Marks 3+2+2) Q.3 Explain Meselson–Stahl experiment and results to prove that DNA replication is semi conservative. Also draw the diagram. — OR (5+2) — Explain how the movement of protons across mitochondrial membrane is involved in ATP synthesis. Also draw diagram showing movement of electrons.

Q4. (Marks 5+2) Q.4 Describe Nitrogen Fixation, Ammonification and Nitrification stages involved in Nitrogen cycle. Also draw labelled Nitrogen cycle flow sheet. — OR (4+3) — Explain mechanism of enzyme action through 'induced fit' and 'lock and key' models.

Q5. (Marks 06) Q.5 Epistasis is different from dominance. Explain with the example of coat color in Labrador retriever. — OR (2x3) — Identify and explain the properties of water which enable it to: a) Act as temperature stabilizer b) Help organisms to live in warm environment c) Help organisms to live under ice

Q6. (Marks 4+2) Q.6 Describe the components of theory explaining the ascent of sap in plants. Also draw the diagram. — OR (2x3) — Write three distinguishing characteristics and one example of each of the phylum Porifera, Platyhelminthes and Annelida.